

Paper Directions 12 — Guidance for Charles

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Source: `transcripts/20260520_Paper_Directions_12_otter_ai_transcript.docx` (Otter; ~16 min)

Interpretation: `Alex_model_interpreted.md` § PD12 addendum

Related: PD11 (pool pre-sorting, May 15) — separate meeting; three threads, not four

Context

Near the start of PD12, Alex confirmed his theoretical memo (`Alex_model.md`) was produced in ~45 minutes with AI assistance from prompts. Treat it as a **structured brief and mechanism checklist**, not finalized paper prose. **PD12 overrides or sharpens** several terms — especially **capability** (own performance / differentiation within peer group, not institutional strength) and **crowding vs. quality**.

Alex separated **four priority workstreams** in the second half of the call. They are **not the same task**.

Priority 1 — Parameter identifiability (simulation + fit)

Alex's ask: Fit the generative model (~5–6 parameters) to **real data in all three domains** currently in scope (MBB, tenure, ...).

Concern: The model may be **overparameterized** — many parameter combinations may fit similarly.

Remedies discussed:

- **Fix** some generative knobs across worlds (e.g. same Beta hyperparameters everywhere — not estimated per dataset).
- **Roll** mechanism-specific terms into a single outcome Y per domain ("in this dataset we roll a, b, c into Y ; in that one, a, b, d ").
- **Lower dimensionality** until recovered parameters are **unique and interpretable**, not just curve-fitting inverted-U shapes.

Distinction: Identifiability = "Can we estimate the structural parameters of *this* model from *these* tables?"

Repo hooks: `539_alex_model.ipynb`; formal fit / identifiability sweeps.

Priority 2 — Extreme events (simulation sensitivity)

Alex's ask: Not "rare draft surprises" in data — **knob-turning in the sim:**

- Set $\lambda \rightarrow 0$, turn congestion off, tighten viability, etc.
- Confirm the inverted-U and mediation patterns **disappear mechanistically** when they should.

Purpose: Validate the **internal logic** of `539_alex_model.ipynb` before over-interpreting one empirical ventile plot.

Repo hooks: `539_alex_model.ipynb` parameter kill switches; compare to CELL 10 ability-only null (talent-only fails).

Priority 3 — Model-guided empirical features

Alex’s ask: The theory should **propose new measurements**, not only fit old ones.

Flagship example:

- Explicit **congestion** in MBB — quantify $C_{i,t}$.
- Test whether congestion **mediates** the downturn in $\bar{a}_t$.
- Kentucky/Duke mega-recruit years as **quasi-experiments**.

Charles’s clarification (~12:15): Ideal congestion measure **indirectly reflects** talent in the system but **does not enter the outcome as raw** a_i — study “how crowded is this team?” and predict harm to marginal members. In practice θ is built from a_j or prospect signals (Y_{draft} , combine, recruiting); document transparently.

Charles’s commit after the call (~8:22): Focus on adding the **congestion crowding term** in the empirical pipeline (not replacing all of 538, but making $C_{i,t}$ first-class).

Key conceptual split (PD12):

Quantity	Role
Team quality $\bar{a}_t$	Mean peer performance — <code>poolq_loo</code> / LOO pool quality
Viable-peer congestion $C_{i,t}$	Density of teammates above prospect threshold — <code>crowding_smooth</code> , viable-peer counts

Repo status (June 2026):

- `crowding_smooth` / `pool_c_smooth_loo` — implemented in `tier1_pool_assignment.py` + 538D CELL 10
- Near-threshold heterogeneity — CELL 4D exports (`heterogeneity_ventiles_top_tail.*`)
- Do **not** treat LOO **sum** as crowding at fixed roster size (redundant with mean)

Natural experiments Alex named: Kentucky/Duke mega-recruiting years; transfer portal/NIL; cohort shock / DiD; mobility reversal (easier in tenure than MBB single-season panel).

Priority 4 — Falsification and scope

Alex’s ask: The three main systems were **chosen** because they plausibly satisfy the model’s conditions. A strong paper also shows **where it fails**.

Tasks:

- Find a **fourth or fifth setting** where the inverted-U **does not** appear (NHL draft mentioned).
- Explain **which assumption breaks** (e.g. different attention structure, evaluation noise, no local peer congestion for slots).
- Explore **new positive domains** if data exist — Charles suggested **macro economics** (GDP, patents, oil production vs. GDP across nations competing for firms). Alex: “really nice level of estimation” at country–company scale.

Timeline: “Let seep over weeks” — not immediate.

Critical PD12 clarifications (not the four priorities, but binding)

Capability vs. substitutability

Misread	Corrected read (Alex, PD12)
Organizational/program capability (e.g. “Duke’s institutional strength”)	Own performance and differentiation within the local peer group — proxy for latent talent a_i

In elite environments everyone is strong but **substitutable**; identifiability collapses even when absolute a_i is high.

Army CIF figure (Charles showed)

- **Promotion (top):** Inverted-U with elite-tier dip — cross-domain shape exists.
- **Attrition (bottom):** Not a mirror image by construction; late-bin rebound (“superstar tail”) visible — interpret with care (few people / wide uncertainty in top bins).
- PD12 point: **shape exists across domains**, not that promotion and attrition must move opposite bin-for-bin.

One-paragraph paper sentence (PD12-aligned)

We distinguish **team quality** (mean peer performance, $\bar{a}_t$) from **viable-peer congestion** (density of teammates above a prospect threshold, $C_{i,t}$). Elite environments raise both average talent and **substitutability**; the inverted-U in advancement is predicted to steepen on the right primarily where $C_{i,t}$ is high, especially for **near-threshold** individuals, while **top latent talent** still clears selection — consistent with the small recovery at the highest ventiles in our NCAA draft data.

Suggested implementation order (from Alex_model_interpreted)

1. `tier1_mechanism_vars.py` / 530 export: `peer_viable_count_loo` (and optional smooth $C_{i,t}$ from 539 logic); keep `poolq_loo` as quality leg.
2. 538 / 538D CELL 10: `crowding_smooth` mode live; document quality vs pool-mean axis (Plot B toggle).
3. `539_alex_model.ipynb` : Cells for Priority 1–2 (fit / identifiability, parameter kill switches).
4. `panel_rebuild` / `combine_bridge` : Optional `Y_combine_meas` , `Y_nba_signal` on panel for θ robustness.
5. **Writing:** Short PD12 cross-ref in advisor briefs; **do not** claim all six mechanisms are separately identified in MBB v1.

Mapping to v1 manuscript plan (June 2026 agent consensus)

PD12 priority	v1 manuscript status (Path II)
1 Identifiability	Deferred — qualitative generative POC; not full 3-domain calibration
2 Extreme events	Partial — talent-only fails (CELL 10); formal 539 kill-switch sweeps not draft-critical
3 Congestion features	Primary alignment — crowding_smooth, CELL 10, 4D heterogeneity; SCOUT D10 bundle next
4 Falsification	Deferred — NHL / 4th domain not in Summer–Fall 2026 scope

Tension to manage with Alex: PD12 is richer than the locked v1 path. Agent correspondence (Rounds 1–5) parked 1 and 4 to enable manuscript draft. Priority 3 is what current SCOUT/VECTOR work delivers.

Open questions left on the table (PD12)

- Exact θ for MBB: global PPM quantile vs combine/draft-based $Y_{\text{nba_signal}}$ vs smooth logistic viability (539).
- Whether to **mediate** with $C_{i,t}$ in LPM only or also in Tier 1 generative selection (538).
- NHL (or other) **negative case** data feasibility for Priority 4.

End PD12 guidance.